

On the Need for Mixed Media in Distributed Requirements Negotiations

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Abstract—Achieving agreement with respect to software requirements is a collaborative process that traditionally relies on same-time, same-place interactions. As the trend toward geographically distributed software development continues, colocated meetings are becoming increasingly problematic. Our research investigates the impact of computer-mediated communication on the performance of distributed client/developer teams involved in the collaborative development of a requirements specification. Drawing on media-selection theories, we posit that a combination of lean and rich media is needed for an effective process of requirements negotiations when stakeholders are geographically dispersed. In this paper, we present an empirical study that investigates the performance of six educational global project teams involved in a negotiation process using both asynchronous text-based and synchronous videoconferencing-based communication modes. The findings indicate that requirement negotiations were more effective when the groups conducted asynchronous structured discussions of requirement issues prior to the synchronous negotiation meeting. Asynchronous discussions were useful in resolving issues related to uncertainty in requirements, thus allowing synchronous negotiations to focus more on removing ambiguities in the requirements.

Index Terms—Global software development, requirement negotiation, computer-mediated communication, media selection, empirical study.

1 INTRODUCTION

SOFTWARE development is a collaborative problem-solving activity where success depends on the ability to create, share, and integrate information [47]. Defining and specifying often large sets of functional and nonfunctional system properties is an intense collaborative process and, in particular, a negotiation process that requires developers and clients to develop a common understanding of the problem, as well as of the solution space [28]. Although research in requirements engineering reports on the inherent communication gap between the different groups of stakeholders [33], the diversity of goals and conflicts in the priorities of the stakeholders involved lead to additional challenges in agreeing requirements. Collaborative software-engineering processes such as WinWin [4], [5] include iterative cycles of identification of the stakeholders' win conditions followed by an exploration and negotiation of mutual agreements. This is where the problem lies at present, given the current trends toward globalization in the software development industry. The need for collaboration (including real-time communication) creates an additional challenge to effective requirements negotiations. As companies turn to outsourcing as a business model (see, for example, [13]), facilitating effective collaboration across

distances becomes problematic. In-depth investigations of collaboration and requirements management processes in multisite organizations (for example, [14]) have identified significant challenges faced by distributed teams. Communication between globally distributed clients and developers is problematic because of differences in software processes, professional backgrounds, and national and organizational cultures among the sites. Failure to achieve a common understanding of system features, combined with a reduced trust and the inability to effectively resolve conflicts, result in budget and schedule overruns and, ultimately, in a damaged client-supplier relationship.

In this study, we examined distributed software teams and the need for mixed communication media in supporting requirements negotiations. This negotiation is more effective when the participants share a mutual understanding, also referred to as common ground in communication. The common ground is the knowledge that participants have in common when communicating and the awareness of it [8]. Not necessarily based on previous knowledge, a common ground is usually dynamically established through an interactive process, called grounding, through which participants exchange evidence about what they do or do not understand over the course of a conversation and interpret cues obtained during social interaction. However, one cannot assume a common ground at the beginning of a software project and we studied ways in which mixed communication media can facilitate grounding and then make distributed requirements negotiations more effective. Specifically, we were interested in investigating appropriate computer-mediated support for distributed software inspections and requirements workshops, activities that have been traditionally recognized as facilitating the establishment of a common

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ground in software development [17], [35]. Being collaboration intensive, these activities largely rely on same-time, same-place interactions of the project stakeholders and, thus, there is a need to further our understanding of how to achieve a shared understanding among distributed teams that do not benefit from rich communication media such as face-to-face interactions. Although there are studies of software inspections for validation of requirements negotiation models [18], [19], there is scarce research on how inspections can enable effective negotiations in distributed teams. Although distributed negotiations that bring together project stakeholders in real-time conversations are difficult and expensive to coordinate, the existing literature [10], [11] indicates that real-time (that is, synchronous) communication is the best fit when it comes to group performance. The extensive interactivity enabled by rich media such as synchronous communication allows the resolution of ambiguities in requirements, whereas leaner media (asynchronous text-based media) is regarded as nonconductive to negotiations but, rather, may be useful for clarifying issues not requiring negotiation.

As such, in this paper, we claim that there is a need for mixed media in supporting effective distributed requirements negotiations, contrary to the traditional belief that synchronous meetings alone are the most effective communication medium for requirements negotiations. Presuming that there will always be a need to hold synchronous requirements negotiations, we have investigated ways to increase the effectiveness of real-time requirements workshops by means of asynchronous discussions as part of distributed inspection processes. Our findings indicate that the requirements negotiations were more effective when the groups conducted asynchronous discussions prior to the synchronous negotiation meetings.

The paper is structured as follows: In Section 2, theories related to computer-mediated communication and the effects of media selection on the effectiveness of collaboration are discussed, as well as our study hypotheses that were informed by these theories. In Section 3, we describe the context of the study, the research design, and a detailed report of study results. In Section 4, we discuss these results in relation to our research hypotheses, as well as threats to validity. Finally, we conclude in Section 5 with implications for research and practice.

2 THEORETICAL BACKGROUND AND DEVELOPMENT OF HYPOTHESES

The study of grounding in distributed, computer-mediated environments has to be related to media-selection theories. Clark and Brennan posit that people should attempt to achieve a common ground with techniques available in a communication medium leading to the least collaborative effort [8]. Experiments suggest that people who have little common ground profit from video and audio-based communication channels for completing collaborative tasks [46], whereas people who have an extensive preexisting common ground can communicate effectively over text-based single-channel media such as e-mail [34].

Our central thesis is that a mix of synchronous and asynchronous communication media is needed to achieve common ground in distributed requirements negotiations in an effective manner. In an effort to create rich media environments that allow exploration of multiple viewpoints in requirements meetings, videoconferencing has been proposed as a medium to facilitate distributed requirements negotiations. Although a large body of literature debates the usefulness of videoconferencing over audio conferencing for distributed group work, there is some consensus that the addition of video is valuable in negotiation situations and relationship building (for example, [1]). A study of distributed requirements negotiations found videoconferencing-supported meetings to be as conducive to win-win situations as face-to-face meetings [12]. However, there are theoretical, as well as pragmatic, considerations for which videoconferencing (and synchronous medium in general) alone is not the most desirable medium for effective distributed requirements negotiations. First, the practicality of organizing videoconferencing sessions is low as synchronous interaction and videoconferencing sessions in particular have additional overhead. The necessary infrastructure is expensive to set up and maintain at remote sites and its coordination across organizational boundaries is often problematic.

Second, media selection theories from the literature suggest that relying on synchronous media alone may not yield the best results in terms of performance and that a combination with a less rich media (for example, other than face-to-face) would facilitate a more rational approach to decision making. These theories form the theoretical basis for our empirical investigation and have informed the development of our study hypotheses. We review the relevant media-selection theories in the next section, after which we describe our hypotheses in detail.

2.1 Media-Selection Theories

The media richness theory [10], [11] argues that performance depends on the appropriateness of the match between the media characteristics and information requirements of the task (clarification versus additional information) and that matching media to collaborative tasks is motivated by the need to reduce uncertainty and equivocality. Uncertainty is defined as *the difference between the amount of information required and the amount of information already possessed about a situation*, while equivocality is the *existence of multiple and conflicting interpretations about a situation* [11].

Thus, media can be ranked according to their richness, with face-to-face interaction as the richest, followed by videoconferencing, and written text as the leanest medium. The media richness theory asserts that rich media would lead to better group performance for equivocal communication situations, while lean media are suited for uncertain communication situations. Equivocality is reduced by providing sufficient clarifications and this is favored by the capacity of rich media to enable users to explore the multiple interpretations of the situation by communicating more quickly and exchanging more types of information. On the other hand, uncertainty can be reduced by obtaining sufficient additional information—a matter of

searching for a clearly identifiable answer—possibly with lean media.

More recently, however, the media switching theory [38] argues that this is too simplistic to explain the ways in which the choice of media influences the cognitive processes underlying communication: In fact, a variety of media is needed for complex decision-making tasks where information overload is a likely risk. This theory recognizes an interesting paradox of richness and calls for the need to balance attention and motivation with the ability to process information.

On the one hand, face-to-face meetings and, more generally, synchronous communication media, as typical examples of high social presence media (in which one perceives the presence of participants in the communication, including verbal and nonverbal cues [42]), require participants to be present at the same time, if not at the same place. This, in turn, requires a high level of commitment to the communication process, further providing the ability to reciprocally monitor attention. In contrast, e-mail communication and asynchronous communication in general (lower social presence) must compete with other activities on the receiver's side and can be easily ignored with no feedback to the sender. As a result, high social presence media ensure that motivation and attention is maintained, while the opposite is true for low social presence media.

On the other hand, low social presence media allow a receiver to gain time to think at their will, as well as being able to find additional information sources, until comprehension is fully achieved. On the contrary, when faced with highly complex messages sent with a high social presence, the receiver could be overwhelmed with information, thus delaying or biasing his or her decision. As a result, low social presence media provides a greater ability to process information than high social presence media.

In conclusion, the use of rich media high in social presence should be used to assure attention for small amounts of information. The use of lean media low in social presence, although possibly resulting in decreased motivation, increases the ability to process large amounts of information over longer periods of time.

2.2 Development of Hypotheses: A Mix of Lean and Rich Communication Media to Support Distributed Requirements Negotiations

To review our central thesis, we propose that a mix of lean and rich media is needed to improve requirements engineering activities when stakeholders are geographically dispersed. In particular, although rich media high in social presence—such as synchronous videoconferencing meetings—are needed for removing equivocality and converging to a shared agreement, lean media low in social presence—such as asynchronous text-based discussions—are valuable in providing an early mechanism to remove uncertainties, that is, to clarify issues, and thus provide a better focus and structure to the discussion during synchronous negotiation sessions. According to the media switching theory [38], even though synchronous videoconferencing meetings may ensure project stakeholders' motivation and attention in the discussion of possibly conflicting

requirements, the high social presence important to support social relationships could also impede unbiased or prompt decisions. Therefore, an asynchronous text-based communication medium emerges as a useful complement for the preparation of such meetings: They allow the group of participants to process information and consider requirements issues and provide missing information at their own time and pace.

Hypothesis H1. *Groups of remote stakeholders using rich (synchronous videoconferencing) media for requirements negotiations are more effective in reaching a mutual agreement when they also spend time in text-based (asynchronous) discussions prior to the videoconference.*

Communicating and agreeing on requirements involves constant interplay between reduction of uncertainty and ambiguity in requirements. Activities of requirement gathering are typically followed by modeling and inspection activities which trigger the need to further collect information about requirements and their context (reducing uncertainty) and resolve conflicts due to multiple viewpoints in the requirements engineering process (reducing equivocality).

According to the media richness theory [10], [11], we thus posit that a preliminary resolution of uncertainties through an asynchronous discussion prior to the synchronous negotiation can shorten the list of requirements issues to be negotiated during real-time discussions. By means of asynchronous communication, participants can take their time to collect information that is lacking or answer clients' concerns (when answering involves uncertainty reduction only). There is an opportunity to use the rich media negotiation meetings to mostly resolve ambiguities (equivocality) in requirements since the higher interactivity level available in rich media allows the exploration of multiple viewpoints. Hence, our second hypothesis concerns the role of asynchronous discussions in relation to the participants' ability to remove uncertainty in the requirements engineering communication process (those issues that require clarifications and/or answers without the need to negotiate).

Hypothesis H2. *Asynchronous discussions of requirements issues remove more uncertainties than ambiguities, thus allowing synchronous meetings to focus more on removing ambiguities.*

In addition, a preliminary asynchronous discussion can be useful in anticipating the building of a common ground between remote stakeholders. The text-based time-insensitive nature of asynchronous communications allows for the gathering of information providing context to certain issues in discussions or helps clarify differences in terminology or those issues that do not require negotiation. As a result, participants have the opportunity to start developing a common ground earlier than in synchronous negotiations; there is a greater chance for clarifications and questions to be asked without the added concern of time constraints inherent in real-time requirements meetings. The subsequent synchronous requirements negotiations therefore are expected to more quickly converge to an agreement but with apparently shallow interactions. We expect that real-time conversations

would be rich in references to posted messages exchanged in the asynchronous discussion.

Hypothesis H3. *Mixed-media groups working with asynchronous text-based discussions achieve an early common ground that is useful to negotiate requirements during synchronous videoconference meetings.*

3 EMPIRICAL INVESTIGATION

To investigate these hypotheses, we conducted a case study of six academic teams involved in geographically distributed software projects with an emphasis on requirements engineering activities. As part of the academic course, the project groups negotiated requirements during one-hour synchronous videoconferencing meetings. In our research design, we created two process variants in which half of the groups were involved in asynchronous discussions of open issues prior to the synchronous negotiation meeting, while the other half was not. We were thus able to compare the performance of the groups using a mix of media with that of the groups using only videoconferencing meetings. The details of the course and the process variants of our study design are described as follows:

The six groups observed were involved in a Global Software Development (GSD) course offered at three universities during January-May 2005: the University of Victoria (Canada), the University of Technology (Sydney), and the University of Bari (Italy). The course involved a total of 32 students of whom 12 were master's and doctorate students at the University of Victoria, two were graduate and eight were undergraduate students at the University of Technology, Sydney, and 10 were master's students at the University of Bari.

The course was dedicated to a cross-university student project that was structured as an outsourcing project in which work was allocated to a distributed software group in a different organization (and country). The project outcome was a software requirements specification (SRS) as a negotiated software contract between the software group and the outsourcing company. Teamwork was critical in completing the software project as the software group had to frequently interact with the clients to understand the required software features.

Table 1 shows the student distribution in international teams. The Canadian students worked on software projects with the Australian and Italian groups as follows: the 12 Canadian students formed three groups of four (Gr1-3), the Australian students formed two groups of five (Gr4-5), and the Italian students formed two groups of three and seven (Gr6cl and Gr6dev) students, respectively. Each Canadian and Australian group was involved in two different projects, playing the role of the client (C) and developer (D), respectively. Each of the two Italian groups was involved in only one project, either as a client (Gr6cl) or as a developer (Gr6dev).

3.1 The Student Projects and Their Requirements Engineering Process

There were three distinct projects, each with two instances. Project A (A1 and A2 in Table 1) was to design a GSD system to facilitate GSD collaboration. In project B (B1 and B2), the students designed the interface for an "iMedia" software to allow users to purchase movies online and

TABLE 1
Groups and Allocation to Course Projects

Country	Group	Project A (A1, A2)		Project B (B1, B2)		Project C (C1, C2)	
		PT1	PT2	PT3	PT4	PT5	PT6
Ca	Gr1	Client (C)					D
	Gr2		D	C			
	Gr3				D	C	
Au	Gr4	Developer (D)			C		
	Gr5		C			D	
It	Gr6cl						C
	Gr6dev			D			

organize and play their movies. Finally, project C (C1 and C2) involved the design of a real-estate system.

The goal of each project team was to develop a requirements specification (RS) in an iterative process. Reaching a mutual understanding between clients and developers meant reducing equivocality and uncertainty by resolving all open issues found by clients in an evolving RS throughout the process. Fig. 1 illustrates the RS development life cycle over a period of seven weeks and consisted of 11 phases of continuous requirements discovery and validation, through which the understanding and documentation of requirements had to be improved. Each of these stages consisted of tasks for either the client or developer groups or a team (project) task. The final deliverable, on which the students were graded in the course, was the final version of the RS (RS 2.0), which reflected the shared understanding of the project that the clients and developers built over the 11 phases. The team tasks were supported by synchronous communication (scheduled videoconference sessions) lasting for one hour. The video connection was enabled by Access Grid or Polycom systems, audio connection using Skype, and computer screens sharing via VNC. The 11 phases are the following:

1. *Kickoff meeting.* In this first meeting, students had the opportunity to get to know each other.
2. *Create Request For Proposals (RFP).* Starting with the assigned project topic, the client group created an RFP document that invited developers to propose their solutions to the clients' needs.
3. *Analyze RFP.* In response to the RFP, the developer group had three days in which to analyze it and come up with a list of clarifications that they needed from the clients before proceeding.
4. *Elicit requirements.* The developers and the clients then held a scheduled one-hour requirements elicitation videoconference during which the developers clarified the clients' needs and elicited more requirements.
5. *Create RS 1.0.* A week after the requirements elicitation meeting, the developers delivered an initial RS document (RS 1.0). This document described the features and scope of the project in detail, following the IEEE standards for requirement specifications [20].

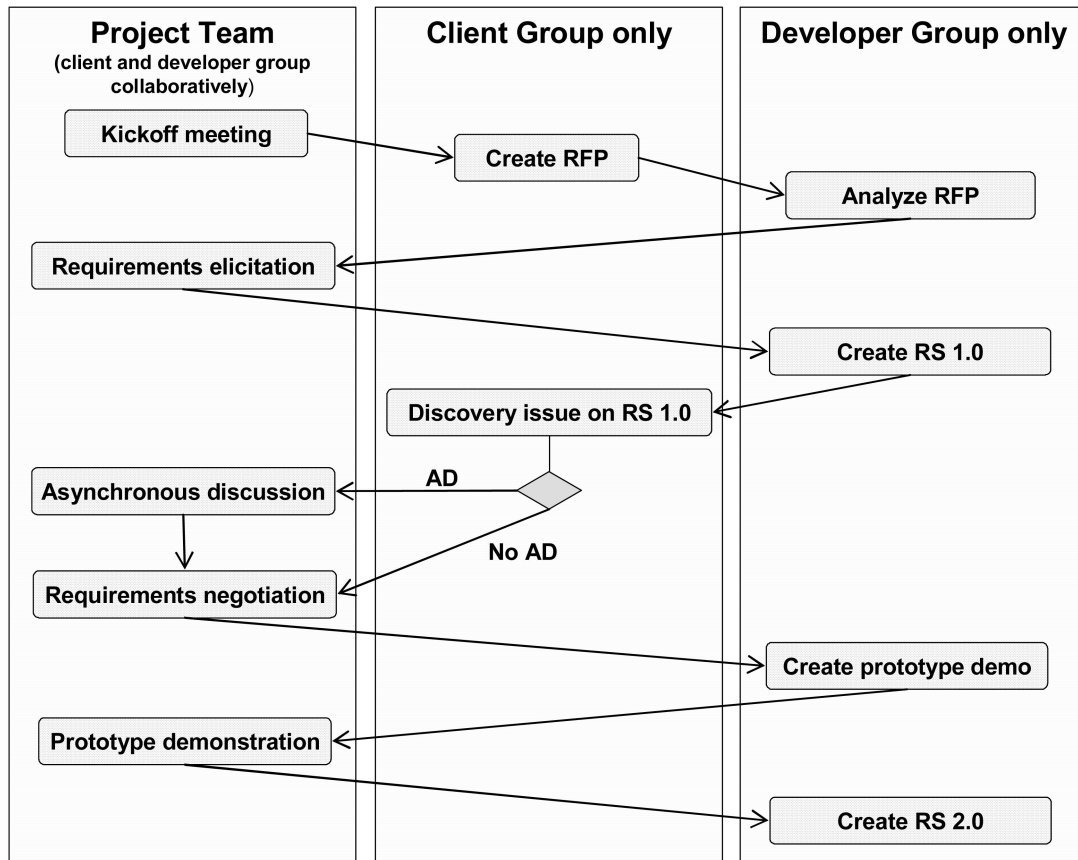


Fig. 1. Two iterative process variants (AD and No-AD) for developing an RS.

6. *Discover issues on RS 1.0.* Upon receiving the RS 1.0 document, the clients had a week to asynchronously carry out an inspection in order to identify gaps in understanding of the requirements. This inspection was performed entirely online through the use of the Internet-Based Inspection System (IBIS) [27]. In this project, the developer group acted as the authors of RS 1.0 and the inspection was carried out by the client group. Each member of the client group individually participated in the Discovery stage of the IBIS tool by reading the RS 1.0 and recording requirements issues in the system. The issue information contained a description of the issue found, as well as a number of issue attributes such as type severity. Each recorded issue was classified according to the IEEE standard taxonomy for good requirements [20], including these categories: omission, ambiguous information, incorrect fact, inconsistent information, not modifiable, and not verifiable. After the Discovery stage, a course assistant performed the Collection stage of the IBIS tool to gather all issues and merge duplicates (that is, issues found by more than one client) into a unique list of collated issues.
7. *Asynchronous discussion.* The entire project team, clients and developers, participated in a four-day asynchronous discussion of the entire list of collated issues. This asynchronous discussion was again

conducted in IBIS (that is, in the Discrimination stage). The purpose of the asynchronous discussion was to reach an understanding of each issue and identify those issues that could be closed online (that is, where resolution could be reached without further negotiation) or remained open issues (everything else, which had to be further negotiated in real-time discussion). The process of closing issues used two mechanisms in IBIS: posting messages with respect to each issue and voting as to whether it is still an open issue or is resolved and thus could be closed.

8. *Requirements negotiation.* Those issues that could not be closed during the asynchronous discussion (that is, remained open issues) were then discussed during a scheduled requirements negotiation held in a one-hour videoconference meeting between developers and clients. Developers acted as the group who led the synchronous discussion. Participants used VNC for application sharing and, in particular, to share the issue list and meeting agenda. The real-time access to the issue list proved useful for recording decisions related to each issue, as well as for displaying the issues that were still open. In preparation for each videoconference meeting, each site (that is, clients and developers) designated one student to play the role of the facilitator and the facilitator from the developer

group proposed the meeting agenda. During the meeting, both facilitators asked participants for feedback and attempted to maintain equality in participation.

9. *Create prototype demo.* After the requirements negotiation, the developer group had one week to develop a prototype of the system to reflect the results of the negotiation. This prototype did not have to contain a working code but could consist of storyboards and paper or computer-based mockups.
10. *Prototype demonstration.* The purpose of this meeting was to express the developers' understanding of the project and their clients' needs, which was done through a one-hour videoconference demo. The clients could give their feedback to the developers and thus reach a consensus on the project between the two groups.
11. *Create RS 2.0.* Finally, three weeks after the prototype demos, the developers submitted a final version of the RS document (RS 2.0). This version incorporated the clients' feedback collected since the first RS draft was written, that is, throughout the requirements negotiation and prototype demo.

The projects were assigned to groups before group membership was determined. The project assignment was performed so that each group worked with a different partner group for each of the two projects to which it was assigned (with the partner group always located in a different country) so that the two projects the group worked on regarded a different topic.

3.2 Study Design

To evaluate our hypotheses, we created two process variants that allowed us to study groups involved in mixed media (asynchronous followed by synchronous negotiations of requirements) or only rich media (synchronous videoconferencing negotiations). We thus manipulated the process variant with two values as the main independent variable (see Fig. 1): 1) the process in which three groups were involved in all stages, including an asynchronous discussion prior to the synchronous videoconferencing negotiation (henceforth referred to as the *AD process variant* or *mixed-media groups*), and 2) the process in which the other three groups did not partake in an asynchronous discussion but, rather, entered the synchronous videoconferencing meeting right after the Discovery stage in IBIS (henceforth referred to as the *No-AD process variant* or *rich media-only groups*). Thus, in order to assess the impact of mixed media on remote requirements negotiation, we compared the performance of the teams in the AD process variant and that of the teams in the No-AD process variant.

For clarification, in the *AD process variant*, both clients and developers used the inspection tool to discover and asynchronously discuss requirements issues. These threaded discussions were stored in the tool. The aim was to come to an understanding of each issue by exchanging messages and to reach an early resolution through a common agreement expressed by voting. Those open issues that could not be closed during the asynchronous discussion were then left for the synchronous requirements negotiation. In the

TABLE 2
Experimental Design

project team (client/developer)	process variant
A1 (gr1/gr4)	No-AD (rich media-only)
B1 (gr2/gr6dev)	No-AD (rich media-only)
C1 (gr3/gr5)	No-AD (rich media-only)
A2 (gr5/gr2)	AD (mixed media)
B2 (gr4/gr3)	AD (mixed media)
C2 (gr6cl/gr1)	AD (mixed media)

No-AD process variant project, the teams skipped the asynchronous discussion and all issues found at the discovery stage were thus considered as open issues to be dealt with at the negotiation. We instructed half of the projects to conduct the asynchronous discussion before the negotiation and the other half of the projects to proceed to the negotiation without asynchronous discussion (shown in Table 2).

3.3 Variables, Measurements, and Results

In this section, we describe the constructs, measurements that we define in order to conceptualize the elements in our hypotheses and the results of our study.

3.3.1 Process Effectiveness

H1. *Groups of remote stakeholders using rich (synchronous videoconferencing) media for requirements negotiations are more effective in reaching a mutual agreement when they also spend time in text-based (asynchronous) discussions prior to the videoconference.*

To conceptualize the elements in our first research hypothesis, our construct for the effectiveness of the synchronous requirements negotiations in reaching a mutual agreement was the number of issues left open after the synchronous negotiation session. Thus, the H1 hypothesis becomes

H1'. *The percentage of open issues after synchronous negotiation is lower in the AD process variant than in the No-AD process variant.*

In order to gain insight into the performance of the six groups in the AD and No-AD process variants, we traced the number of open issues at each stage of the collaborative development process. Thus, we define the dependent variables outlined in Table 3.

Table 4 includes the values of the variables for each of the six projects. The three projects not involved in asynchronous discussions (A1, B1, and C1) entered the synchronous negotiation with different numbers of open issues to be resolved: 40, 61, and 100, respectively. The number of issues that were closed during the synchronous negotiations ranged from 26 to 47, leaving from 12 to 74 issues unresolved. At the same time, an important

TABLE 3
Dependent Variables for Process Effectiveness

Variable	Description
Open issues after discovery	The number of open issues at the end of the discovery stage. It is the number of open issues with which the AD groups entered the asynchronous discussion.
Closed issues during async discussion	The number of issues for which a consensus was reached during the asynchronous discussion. Whatever the agreement, a closed issue did not require any further discussion.
Open issues before sync negotiation	The number of open issues carried over to the synchronous negotiation. In the case of a No-AD process variant this number equals the number of open issues after discovery.
Closed issues during sync negotiation	The number of issues for which an agreement was reached at the videoconference requirements negotiation meeting.
Open issues after sync negotiation	The number of issues for which an agreement was not reached during the synchronous negotiation.
% of open issues after sync negotiation	The ratio of open issues after sync negotiation to open issues after discovery.

difference can be seen in the three projects that conducted asynchronous discussions (A2, B2, and C2); these groups entered the synchronous negotiation with a much lower

number of open issues (12, 12, and 13, respectively) and ended the negotiation with 0, 0, and 3 open issues, respectively.

Similarly, Fig. 2 graphically illustrates the trajectory of open issues throughout the three stages in each of the six projects, indicating how asynchronous discussions improved the effectiveness of synchronous negotiations. It can be seen that all three dotted lines, representing projects in the AD process variant, finished below the three continuous lines (which correspond to projects in the No-AD process variant). Particularly important is project B2, which started with the highest number of open issues after discovery (112), but ended up with a significantly lower number of open issues (that is, three), thanks to the asynchronous discussion.

Fig. 3 provides further details on the discussion of issues during the synchronous negotiations of both process variants. When synchronous negotiations started, participants (clients and developers) had a set of open issues (that is, open issue before sync negotiation) to be discussed and closed. As Fig. 3 shows, all synchronous negotiations of the No-AD process variant ended up with a number of issues that were not discussed at all (that is, 15 percent, 18 percent, and 69 percent) and also with a number of issues that were discussed but not closed (that is, 15 percent, 5 percent, and 5 percent) because the participants did not reach agreement. The same cannot be observed for the groups in the AD process variant, with the exception of the B2 project, where 25 percent of the issues were not discussed.

TABLE 4
Evolution and Resolution of Issues from the Inspection to the Negotiation Phase

project (cl/dev)	open issues after discovery	closed issues during async discussion	open issues before sync negotiation	closed issues during sync negotiation	open issues after sync negotiation	% of open issues after sync negotiation
A1 (gr1/gr4)	40 (uncertainties: 22 ambiguities: 18)	No-AD	40 (uncertainties: 22 ambiguities: 18)	28 (uncertainties: 13 ambiguities: 15)	12 (uncertainties: 9 ambiguities: 3)	30.0%
B1 (gr2/gr6dev)	61 (uncertainties: 22 ambiguities: 39)	No-AD	61 (uncertainties: 22 ambiguities: 39)	47 (uncertainties: 11 ambiguities: 36)	14 (uncertainties: 11 ambiguities: 3)	23.0%
C1 (gr3/gr5)	100 (uncertainties: 55 ambiguities: 45)	No-AD	100 (uncertainties: 55 ambiguities: 45)	26 (uncertainties: 6 ambiguities: 20)	74 (uncertainties: 49 ambiguities: 25)	74.0%
A2 (gr5/gr2)	23 (uncertainties: 17 ambiguities: 6)	11 (uncertainties: 9 ambiguities: 2)	12 (uncertainties: 8 ambiguities: 4)	12 (uncertainties: 8 ambiguities: 4)	0	0.0%
B2 (gr4/gr3)	112 (uncertainties: 84 ambiguities: 28)	100 (uncertainties: 77 ambiguities: 23)	12 (uncertainties: 7 ambiguities: 5)	9 (uncertainties: 5 ambiguities: 4)	3 (uncertainties: 2 ambiguities: 1)	2.5%
C2 (gr6cl/gr1)	23 (uncertainties: 19 ambiguities: 4)	10 (uncertainties: 10 ambiguities: 0)	13 (uncertainties: 9 ambiguities: 4)	13 (uncertainties: 9 ambiguities: 4)	0	0.0%

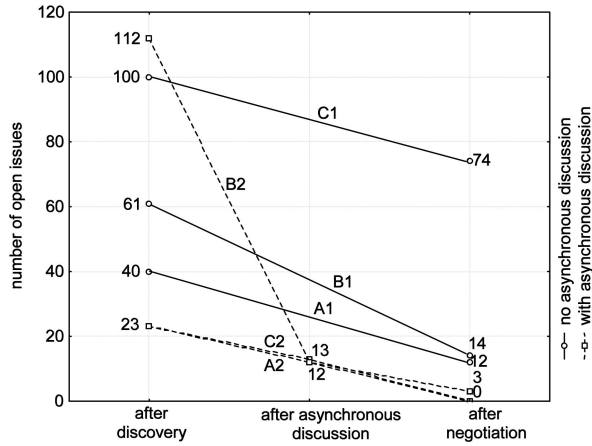


Fig. 2. Number of open issues at the end of the three process stages for the six groups.

To evaluate the H1 hypothesis we compared the percentage of open issues after synchronous negotiation for the groups in the AD and No-AD process variants. We performed the Mann-Whitney U test as a nonparametric alternative to a *t*-test for independent samples [49] because the number of observations is very small (six projects) and we cannot rely on the normality assumption. The test indicates a significant difference at the 5 percent level ($Z = 1.963$ and $p = 0.04$) of the percentage of open issues after synchronous negotiation between the two process variants. Given that the two samples are of size three, we encounter the only case where the Mann-Whitney U test provides a significant difference at the 5 percent level, that is, all values in one sample are greater (or less) than all values in the other [29].

3.3.2 Uncertainty Reduction during Asynchronous Discussions and Equivocality Reduction during Synchronous Negotiations

H2. Asynchronous discussions of requirements issues remove more uncertainties than ambiguities, thus allowing synchronous meetings to focus more on removing ambiguities.

The construct we defined was the type of issues to be discussed during the asynchronous and synchronous discussions. Our intention was to distinguish between elements of uncertainty and equivocality in conversations. We considered that an issue was characterized by *ambiguity* (or equivocality) if it indicates multiple and, possibly, conflicting interpretations of a specific requirement. An *uncertainty* is an issue that indicates the absence of sufficient information in a specific requirement. Thus, we measured the number of *uncertainties* and *ambiguities* in all asynchronous and synchronous discussions. To count *uncertainties* and *ambiguities*, we parsed all issues identified during the IBIS-based discovery stage by clients. We included in the *uncertainty set* all issues classified under the category “omission” of the IEEE taxonomy. Similarly, we included in the *ambiguity set* all issues classified under the category “ambiguous info” of the IEEE checklist. The issues classified in the remaining IEEE categories of “incorrect fact,” “inconsistent info,” “not verifiable,” and “not modifiable” were also analyzed and counted as part of one of the two sets defined above,

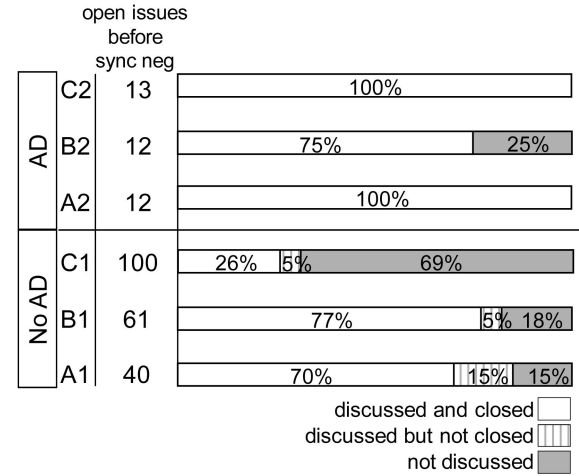


Fig. 3. Distribution of issues that were discussed and closed, issues that were discussed but not closed, and issues that were not discussed at all in all six groups.

depending on whether they required additional information (could be resolved by removing uncertainty and consequently classified in the uncertainty set) or clarifications (meaning was ambiguous and had to be clarified and consequently classified in the ambiguity set). To characterize the conversations in the asynchronous discussions of the mixed-media groups and the synchronous discussions of both mixed-media and rich media-only groups, we defined the dependent variables outlined in Table 5.

Thus, the H2 hypothesis was reformulated into the following two hypotheses:

H2a. During the asynchronous discussions of the mixed-media groups, the percentages of closed uncertainties are higher than the percentages of closed ambiguities.

H2b. During the synchronous negotiations of all groups, the percentages of closed ambiguities are higher than the percentages of closed uncertainties.

Table 6 shows the values of these variables for the six groups in the two process variants. One can observe that asynchronous discussions were more useful to close uncertainties than ambiguities. The percentages of closed uncertainties during asynchronous discussions (0.53 percent, 0.91 percent, and 0.53 percent, respectively for A2, B2, and C2) were always higher than the percentages of closed ambiguities during asynchronous discussions (0.33 percent, 0.82 percent, and 0.0 percent, respectively for A2, B2, and C2) for all three projects of the AD process variant. Although participants had many uncertainties to be discussed during the asynchronous discussion, they were able to close many of them. Similarly, higher percentages of closed ambiguities than closed uncertainties during synchronous negotiations were recorded for each of the six projects.

To test for statistical significance, we performed two Wilcoxon matched-pair tests as a nonparametric alternative *t*-test for dependent samples [49] and found that both differences were significant at the 10 percent level (with $Z = 1.603$ and $p = 0.10$ and with $Z = 1.603$ and $p = 0.10$, respectively).

TABLE 5
Dependent Variables for Equivocality and Uncertainty Reduction

Variable	Description	Hypothesis
Uncertainties after discovery	The number of open issues that require additional information other than clarifications to be resolved, and classified as uncertainties	H2a
Ambiguities after discovery	The number of open issues that require clarifications about the existing information to be resolved, and classified as ambiguities	H2a
Closed uncertainties during async discussion	The number of uncertainties after discovery for which consensus was reached during the asynchronous discussion.	H2a
Closed ambiguities during async discussion	The number of ambiguities after discovery for which consensus was reached during the asynchronous discussion.	H2a
% closed uncertainties during async discussion	The ratio of closed uncertainties during async discussion to uncertainties after discovery.	H2a
% closed ambiguities during async discussion	The ratio of closed ambiguities during async discussion to ambiguities after discovery.	H2a
Uncertainties before sync negotiation	The number of open issues classified as uncertainties before starting synchronous negotiations (in the case of rich media-only groups this variable has the same value of uncertainties after discovery).	H2b
Ambiguities before sync negotiation	The number of open issues classified as ambiguities before starting synchronous negotiations (in the case of the No-AD groups this variable has the same value of ambiguities after discovery).	H2b
Closed uncertainties during sync negotiation	The number of uncertainties before sync negotiations for which a consensus was reached during the synchronous negotiation.	H2b
Closed ambiguities during sync negotiation	The number of ambiguities before sync negotiations for which a consensus was reached during the synchronous negotiation.	H2b
% closed uncertainties during sync negotiation	The ratio of closed uncertainties after sync negotiations to uncertainties before the sync negotiation.	H2b
% closed ambiguities during sync negotiation	The ratio of closed ambiguities after sync negotiation to ambiguities before the sync negotiation.	H2b

3.3.3 Common Ground in Synchronous Negotiations

H3. *Mixed-media groups working with asynchronous text-based discussions achieve an early common ground that is useful to negotiate requirements during synchronous videoconference meetings.*

We sought to gain insights concerning conversations during the synchronous negotiations and, in particular, whether a greater common ground could be identified in the AD process than in the No-AD process. For this reason, we chose three ways in which to observe and conceptualize the presence of a common ground in the synchronous negotiations and the fact that the groups benefited from it when closing issues. Hence, we split the H3 hypothesis into five hypotheses and also collected additional information that did not follow a particular hypothesis but helped to understand whether groups achieved a common ground.

First, greater levels of common ground would be indicated by fewer speaking turns and spoken words per turn, as some level of understanding in the conversation would eliminate the need to discuss an issue in depth [41], [9], [31], [21]. Thus, we first conceptualized a common ground by conversation efficiency and the constructs we defined were speaking turns per issue and spoken words per turn. The first two hypotheses, H3a and H3b, are given as follows:

H3a. *Mixed-media groups use fewer speaking turns per issue to reach an agreement than rich media-only groups.*

H3b. *Mixed-media groups use fewer spoken words per turn to reach an agreement than rich media-only groups.*

We defined the variables outlined in Table 7. To quantify these variables, an analysis of the transcripts of the

TABLE 6
Uncertainty and Equivocality Reduction

	No AD			AD		
	A1	B1	C1	A2	B2	C2
% closed uncertainties during async discussion	-	-	-	0.53	0.92	0.53
% closed ambiguities during async discussion	-	-	-	0.33	0.82	0.00
% closed uncertainties during sync negotiation	0.59	0.50	0.11	1.00	0.71	1.00
% closed ambiguities during sync negotiation	0.83	0.92	0.44	1.00	0.80	1.00

TABLE 7
Dependent Variables for Conversation Efficiency

Variable	Description	Hypothesis
Discussed issues	The number of open issues discussed during the synchronous negotiation.	H3a
Speaking turns	The number of turns (or continuous speech by one participant) in the oral conversation.	H3a / H3b
Speaking turns per issue	The ratio of speaking turns to discussed issues.	H3a
Spoken words	The number of words spoken by participants during the synchronous negotiation.	H3b
Spoken words per turn	The ratio of spoken words to speaking turns.	H3b

videoconferencing synchronous discussions was performed and the unit of analysis was the issue, as identified during the asynchronous discussion in IBIS. In this analysis, a count of speaking turns and spoken words was recorded for each issue.

Figs. 4 and 5 provide actual values for the variables *speaking turns* and *spoken words per turn* in the synchronous conversations of the six groups. One can observe that, for those projects in which negotiations were preceded by an asynchronous discussion (that is, A2, B2, and C2), the number of *speaking turns per issue* (that is, 11.08, 8.83, and 7.92, respectively) and *spoken words per turn* (that is, 35.09, 24.83, and 14.96, respectively) were higher than the projects in the No-AD process variant, although the two Mann-Whitney U tests on the H3a and H3b hypotheses indicated no significant differences (with $Z = 1.527$ and $p = 0.12$ and with $Z = 0.218$ and $p = 0.82$, respectively).

Second, we expected that the presence of a common ground would be indicated by fewer *questions* and *explanations* for an issue as some understanding of the issue could be assumed from the asynchronous discussion. To investigate the presence of questions and explanations related to issues in the conversation, we performed a content analysis on the transcripts of the video-recorded synchronous negotiations.

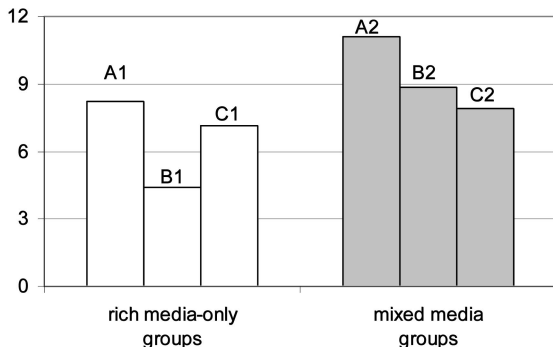


Fig. 4. Speaking turns per issue during synchronous negotiation by process variant.

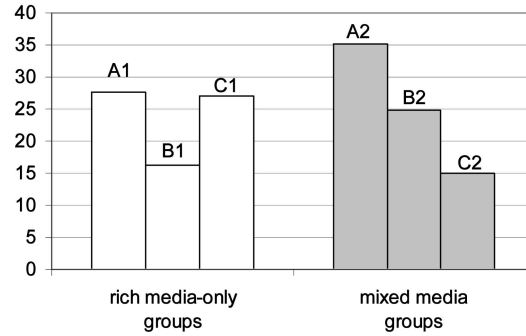


Fig. 5. Spoken words per turn during synchronous negotiation by process variant.

Content analysis, also called coding [40], [45], is a mix of quantitative and qualitative data [40], [45], [24], [43], [48], [32] because it transforms qualitative data (that is, free text, either spoken or written) into quantitative data (that is, numbers). In fact, unlike qualitative research, in content analysis, we have a finite set of possible categories, so the analysis may be performed in a more readily quantifiable manner.

In order to perform this analysis, a coding schema required definition. Some coding schemas have been proposed in the literature [16], [24], [6], [23], but none fully satisfied our needs. From a survey of computer-mediated communication content analysis studies, Rourke et al. [39] found that the unit of analysis more frequently used was the thematic unit introduced by Henri [25] and defined as *a single thought unit or idea unit that conveys a single item of information extracted from a segment of content*. The thematic unit was found to be more satisfactory and one of the researchers identified the units of the transcript to be recorded and categorized from the transcripts of the negotiation conversations.

The categories (thematic units) we defined are listed in Table 8. Among these six categories, *questions* and *explanations* (further categorized as *extra info* or *clarifications*) relate to the common ground, as the fewer the questions and explanations, the higher the level of common ground in the conversation can be assumed. Thus, the last three hypotheses, H3c, H3d, and H3e, are given as follows:

H3c. *Mixed-media groups use fewer questions to reach a consensus than rich media-only groups.*

H3d. *Mixed-media groups use fewer clarifications to reach a consensus than rich media-only groups.*

H3e. *Mixed-media groups use less extra info to reach a consensus than rich media-only groups.*

Two coders performed the content analysis separately and the intercoder agreement was measured by Cohen's kappa and ranged from 0.84 (for project A2) to 0.94 (for project A1). Table 9 shows the details of the conversations in each of the six groups, namely, the number of discussed issues and identified thematic units, as well as the breakdown of these thematic units per categories, that is, the ratio of each of the categories to the total number of issues. For example, there were 2.12 thematic units classified as agreements per issue in group A1's conversations. Our interest was to observe any differences between the numbers recorded for the rich media-only versus mixed-media groups. One can observe the

TABLE 8
Categories of the Adopted Coding Schema

Category		Description
EXPLANATION	QUESTION	Asks the other group or participant any question regarding new or unmentioned information about the issue.
	EXTRA INFO	Adds new information about the issue that has not been elicited by the other group. It can be also an answer to a previous question asked by the other group. It does not include any form of agreement or disagreement expression, but it just provides extra information about the issue. It is a category specific for issues classified as uncertainties.
	CLARIFICATION	States a clarification without adding new information about the issue. It can be also an answer to a previous question asked by the other group. It does not include any form of agreement or disagreement expression, but it just provides a clarification about the information already contained in the issue description or in the messages posted during asynchronous discussion. It is a category for issues classified both as uncertainties and ambiguities.
	AGREEMENT	States an agreement with the issue found or something said in a previous turn or in a message written in the discussion forum by the other group. It could be also an answer to a previous question asked by the other group.
	DISAGREEMENT	States a disagreement with the issue found or something said in a previous turn or in a message written in the discussion forum by the other group. It could be also an answer to a previous question asked by the other group.
	OTHER	Any other kind of communications and tasks, it includes also social communications and technical problems that occur during the video conference negotiation.

following statistically significant results for the H3c, H3d, and H3e hypotheses:

1. The mixed-media groups had a greater number of *questions* ($Z = 1.963$ and $p = 0.04$) and *clarifications* ($Z = 1.963$ and $p = 0.04$) per issue than the rich-media-only groups.
2. The amount of *extra info* (provided to close an issue) per issue was consistently lower for the mixed-media groups ($Z = 1.963$ and $p = 0.04$).

TABLE 9
Results of Applying the Coding Schema to Transcripts of Synchronous Negotiations

	No AD			AD		
	A1	B1	C1	A2	B2	C2
Discussed issues	34	50	31	12	12	13
thematic units	350	245	298	174	141	125
<i>questions</i> per issue	1.50	0.62	1.29	3.33	2.83	2.31
<i>extra info</i> per uncertainties	1.81	1.09	2.00	0.63	0.86	0.44
<i>clarifications</i> per issue	3.03	1.66	3.32	5.42	4.67	3.62
<i>agreements</i> per issue	2.12	1.36	3.00	2.83	2.92	2.69
<i>disagreements</i> per issue	1.09	0.26	0.32	1.42	0.42	0.08
<i>others</i> per issue	1.71	0.76	1.10	1.08	0.42	0.62

Statistically significant results at the 5 percent level are indicated in bold.

Third and last, to investigate the presence of a common ground as an aid in decision making, which characterizes requirements negotiations, we sought to identify whether synchronous conversations made references to elements from the asynchronous discussions.

We conducted a referential analysis as follows: *Referring* is a collaborative process because it involves the speaker who proposes a referring expression and the hearer who accepts or rejects the referring expression as adequate for identifying the referent. Reference, as an aspect of the conversation, has been mostly studied by psychologists [7], [22] and these studies have shown that the first time an item is mentioned, the speaker has to provide a lengthy description before the hearer can identify it correctly. When this process is repeated over several trials, however, the number of words used to describe the item decreases. As previous references to the item have been established as being mutually understood and, hence, part of participants' common ground, speakers can rely on the acceptance of a previous description and use a shorter referring expression based on the earlier description [7], [22].

In our case, the referred objects were messages exchanged during the asynchronous discussions and posted in the IBIS tool. We sought to understand how many times *posted messages* were *referred* to during the synchronous negotiation and whether they were *useful to close an issue* in the synchronous discussions. In our analysis, we did not need to distinguish between the first occurrence of a referred posted message and the following ones within the same speaking turn.

Participants in the AD process variant synchronously negotiated with the help of a document, shared via VNC, which included all posted messages for each issue as recorded in the IBIS repository (referred to as *posted*

TABLE 10
Dependent Variables for the Referential Analysis

Variable	Description
References to posted messages	The total number of references to posted messages (both about uncertainties and ambiguities) that were made during the synchronous negotiations.
References to posted messages about uncertainties	The number of references to posted messages about uncertainties that were made during the synchronous negotiations.
References to posted messages about ambiguities	The number of references to posted messages about ambiguities that were made during the synchronous negotiations.
Discussed uncertainties	The number of open issues before sync negotiation classified as uncertainties which were at least discussed during the synchronous negotiations.
Discussed ambiguities	The number of open issues before sync negotiation classified as ambiguities which were at least discussed during the synchronous negotiations.
References to posted messages per discussed issue	The ratio of references to posted messages to discussed issues.
References to posted messages per uncertainty	The ratio of references to posted messages about uncertainties to discussed issues.
References to posted messages per ambiguity	The ratio of references to posted messages about ambiguities to discussed issues.

messages henceforth). In this manner, they could easily refer to a relevant comment reported from a specific participant in the asynchronous discussion. We thus measured the variables shown in Table 10.

We were interested in understanding how many times a posted message written in the asynchronous discussion was useful during synchronous negotiations. Table 11 shows the total number of references to posted messages for each of the mixed-media groups, the total number of discussed issues during the synchronous discussions, and

TABLE 11
Results from the Referential Analysis
of the Conversations in the Mixed-Media Groups

	A2	B2	C2
References to posted messages	29	12	15
References to posted messages about uncertainties / ambiguities	25 / 4	8 / 4	13 / 2
Discussed issues	12	12	13
Discussed uncertainties / ambiguities	8 / 4	7 / 5	9 / 4
References to posted messages per discussed issue	2.4 (=29/12)	1.0 (=12/12)	1.1 (=15/13)
References to posted messages per uncertainty	2.1 (=25/12)	0.7 (=8/12)	1.0 (=13/13)
References to posted messages per ambiguity	0.3 (=4/12)	0.3 (=4/12)	0.1 (=2/13)

the breakdown of references to posted messages that discussed either uncertainties or ambiguities. One can observe that

1. overall, there was at least one reference per issue in each of the three groups, even 2.4 references for group A2, and
2. more references were made to posted messages about uncertainties rather than ambiguities in each of the three groups (2.1 versus 0.3 for A2 and so on).

To exemplify this process of referring, in Tables 12 and 13, we include two excerpts of the transcripts where one or more posted messages were referred to during conversations in the synchronous negotiations. The underlined units are those units that referred to posted messages.

TABLE 12
An Excerpt of the Synchronous Negotiation
of the B2 (gr4/gr3) Project

client:	<i>Well, you've only clarified one issue, which was the voice mail.</i>
client:	<i><u>I seem to remember 6 or 8 different comments or ideas that were mentioned in that part of IBIS for that question. For instance, we don't know if conference calling is allowed. We don't know if you can put someone on hold. I can't remember all the points that were flying to and from there, but we would need to have some expectation of more than just making a call.</u></i>
dev:	<i>What we discussed amongst ourselves and what we believe is that we should be able to put people on hold, and un-hold. And transfer calls. As far as the other issues are concerned such as that of the voice conference, we weren't really sure on that, and we were wondering how important it is for a voice conference.</i>
...	...

TABLE 13
An Excerpt of the Synchronous Negotiation
of the A2 (gr5/gr2) Project

...	...
client:	<i>I think that's what the points we were bringing up in the IBIS discussion, those points that we've actually brought out already in the IBIS discussion.</i>
dev:	<i>Okay, well how about listing those right now and we can go through them point by point.</i>
...	...

4 DISCUSSION

In our study, we investigated the performance of six groups that negotiated requirements during synchronous videoconferencing sessions, three of which benefited from the asynchronous structured discussion of requirements issues using the IBIS inspection tool. We tested the first main hypothesis that groups will be more effective at resolving requirements issues when their synchronous interaction is preceded by the structured asynchronous discussions. We found support for this hypothesis: Groups in the mixed-media process variant were able to end the requirements engineering process with significantly fewer open issues than the groups that were not involved in the asynchronous discussion.

To further our understanding of the factors that contribute to the higher effectiveness of mixed-media groups, we investigated two related hypotheses. Our second hypothesis was that asynchronous discussions provide support for a resolution of requirements, allowing the groups to focus on ambiguities during synchronous negotiations held as videoconference meetings (that is, rich media). Furthermore, this preliminary discussion of issues during the asynchronous discussions would also enable groups to achieve some common ground useful in closing issues during the synchronous meetings, as our third hypothesis stated. We discuss the findings regarding these two hypotheses together in the following paragraphs.

The quantitative data in Table 6 clearly indicates that asynchronous discussions were more focused on reducing uncertainty rather than equivocality, thus supporting our second hypothesis. First, more uncertainties than ambiguities were clarified in the asynchronous discussions. Consequently, when participants had already discussed asynchronously, they began the videoconference negotiation meetings with a shorter list of open issues to be discussed. Second, more ambiguities than uncertainties were resolved during the synchronous discussions of the rich media-only groups. A situation worth mentioning that occurred during negotiations of one of the rich media-only groups was that a number of issues were discussed but not closed. This happened when an issue was too controversial because of its high equivocality; the participants were not able to reach a resolution and proceeded to discuss the next issue without closing the former. Unfortunately, there was not enough time to return to this discussion leaving the

issue unresolved. This suggests that a mix of lean and rich media would have been useful for this group when discussing complex issues: An asynchronous discussion before the synchronous negotiation might have provided some common ground on the issue, thereby speeding up the interactive session. The quantitative evidence supporting this claim is discussed next.

As explained earlier, in our paper, particular importance was dedicated to the study of a common ground and the search for its indications in several forms established during the asynchronous discussions, including 1) the fact that issues discussed but left open during the asynchronous discussion did not require more information in the synchronous discussion and 2) the fact that synchronous meetings of mixed-media groups used references to messages exchanged during the asynchronous discussions. In this direction, the results of our content analysis (see Table 9) indicate that a lower number of *extra info* units were necessary for the mixed-media groups (we must recall that the analysis of *extra info* units was made for the issues classified as *uncertainties*). In other words, participants of mixed-media groups did not provide, during the synchronous meetings, additional information for those uncertainties already discussed asynchronously but which remained unresolved. This result is consistent with and can be explained by the findings on the ratios of closed uncertainties and ambiguities during the asynchronous discussions to the total number of open uncertainties and ambiguities recorded before the asynchronous discussions, as shown in Table 6. Since the asynchronous interaction was more conducive to discussing uncertainties, it is understandable that less extra information had to be provided for the uncertainties under discussion in the synchronous meetings.

Similarly, the analysis of references made during the synchronous negotiations (see Table 11) provides interesting but positive results with respect to the presence of a common ground. As hypothesized, we found that all mixed-media groups did make references to messages exchanged during the asynchronous discussion. Not only was at least one reference made per issue in the synchronous discussion, but also more references to posted messages were made about uncertainties than about ambiguities. Of course, this is to be expected since more uncertainties than ambiguities were discussed in the asynchronous interaction, but it further evidences the fact that asynchronous discussions were indeed useful in discussing issues classified as uncertainties.

However, we did not find full support for our third hypothesis. The results on the other variables related to a common ground, such as speaking turns per issue, spoken words per turn, questions, and clarifications, were different from our expectation. Contrary to our expectation that fewer spoken words per turn and speaking turns per issue would characterize synchronous interactions preceded by asynchronous discussions, the participants who had already discussed issues asynchronously did not actually use fewer speaking turns and words during the synchronous negotiations. This is, however, not surprising when one takes into account the number of open issues with which each group started in the synchronous negotiation. As

shown in Fig. 3, the number of open issues to be discussed when negotiations started was extensive for those projects in the No-AD process variant (40 for A1, 61 for B1, and 100 for C1). We believe that the participants did not have the chance to discuss each issue in depth, as was possible for those projects adopting the AD process variant with the same time interval (that is, one hour). It is likely that the participants with fewer issues to discuss did not feel the *time pressure and stress* during the one-hour-long videoconferencing session. Research indicates that time constraints and the number of tasks to be executed are among the factors that affect the ability of individuals to make effective decisions [30], [44], where the time constraint is defined as *the difference between the amount of available time and the amount of time required to resolve a decision task* [2], [37].

Further details concerning this finding emerge from the content analysis. We found, again contrary to our expectations, that participants who had discussed asynchronously prior to the negotiation, exchanged more *questions* and *clarifications* before reaching an agreement about an issue. This is, on one hand, closely related to the conversational efficiency results: Participants in the No-AD process variant were afraid to end the synchronous negotiation with many open issues. Thus, they scanned the long open-issue list (40 for A1, 61 for B1, and 100 for C1) more quickly, without dwelling too much on each issue. On the other hand, the participants in the AD process variant started synchronous negotiations with a lower number of open issues (12 for A2, 12 for B2, and 13 for C2) and thus possibly approached the videoconference meetings with much more tranquility.

4.1 Threats to Validity

A key issue in an empirical study is the validity evaluation of the results. We discuss four groups of threats to validity in our empirical investigation: conclusion, internal, construct, and external threats to validity [50].

Conclusion validity concerns the relation between the treatments and the outcome of the experiment with regard to statistical methods, the reliability of measures, and treatment implementation.

A serious issue which could affect the conclusion validity is how we implemented the No-AD process variant of the rich media-only groups. Although all groups had the same amount of videoconferencing time, the No-AD groups may appear as though they had less time to discuss the requirements issues given the absence of the asynchronous process phase. However, this was in accordance with our goal of studying typical distributed software teams that often are not able to benefit from frequent or long videoconferencing meetings. Therefore, we could not allow more than the scheduled time intervals for the videoconference negotiations for the No-AD groups. The groups in our study were typical of global projects—the large time difference between the groups (Canada has a 9-hour and a 16-hour time difference with Italy and Australia, respectively) allows for very little overlapping time—and, thus, it is reasonable to expect that these groups could not benefit from long meetings; neither could we increase the number of videoconference negotiations to assure a fair comparison between the groups. The process of developing the RS already included four videoconferencing weekly meetings:

kickoff meeting, requirements elicitation, requirements negotiation, and prototype demonstration. We presume that the heterogeneity of groups and organizations typical of distributed projects often makes it difficult to overcome the limitation of one meeting per week. Given the fact that synchronous communication is not possible or practical as much as required to resolve ambiguities during requirements negotiations, the additional asynchronous mode of communication is thus useful to sort through the requirements issues (for example, clarifying uncertainties), in addition to simply providing more time to discuss the requirements.

One might note that two of the three mixed-media groups (A2 and C2) had relatively few issues after the discovery phase to begin with and, thus, our conclusions on the effectiveness of asynchronous discussion hinge on the dramatic reduction in the exceptional group B2 only (112 open issues). However, a similar pattern can also be observed within the No-AD process variant; group C1, which coincidentally had a similar number of open issues after discovery (100), was very different from the other two groups (A1 and B1) in the same treatment.

For the statistical analysis, we used nonparametric tests, which do not rely on any assumed distribution of the underlying data and thus can be valid for even very small sample sizes. Furthermore, we used significance levels of both 0.05 and 0.10 for the statistical tests reported in this paper. Although a 10 percent level makes it more likely that we incorrectly reject the false null hypotheses than inadvertently missing significant differences, the small sample size in this study makes the latter case highly probable. Nevertheless, a 10 percent significance level has been adopted in other empirical software engineering studies [3], [15], [26], [36].

Internal validity concerns rival explanations for the experimental findings that make the cause-effect relationship between independent and dependent variables more difficult to believe. The primary threat to internal validity is the selection effect, which occurs when results can be caused by differences in the participants' characteristics in the groups being compared. Random assignment of subjects to experimental conditions usually reduces this threat, but our experimental design is heavily influenced by the small amount of projects and the restrictions posed by the practical course. There was not much that we could do to mitigate this threat. We were not able to completely randomize the selection and participants' assignment to the different groups, which exhibited notable differences regarding language and background. For example, while Australian and Canadian students were exposed to both levels of the main independent variable, although with different roles (clients or developers), Italian students worked on only one project. Furthermore, English was the first language for the Canadian and Australian students but not the Italians. The Canadians and Australians had the most dissimilar backgrounds; the Canadian groups included eight master's and two doctorate students with good experience in software requirements and development, while the Australian groups included mostly (8 out of 10) undergraduate students. All of the Canadians played

the role of clients in rich media-only groups and this is a confounding factor. However, there is no statistically significant difference between the issues found during the inspection phase of the No-AD and AD process variants, despite the possible effect of this factor.

A second threat to internal validity is instrumentation. An instrumentation effect occurs when differences in the results may be caused by differences in experimental material. Although there were three different project topics in this study, we controlled for any variation by restricting the requests for proposals to being implemented as Web applications.

Construct validity concerns the degree to which both the independent and dependent variables accurately measure the constructs of interest. We measured three main constructs to evaluate our hypotheses: the *Process Effectiveness*, the *Uncertainty and Equivocality Reduction* during asynchronous discussions and negotiation meetings, and *Common Ground* during negotiation meetings. To investigate *Process Effectiveness*, we measured the number of issues that were left unresolved after negotiations in both process variants. Our study groups were asked to consider the synchronous negotiation as a crucial point in finalizing the RS. Thus, we regarded the number of issues left open as an appropriate measure for their performance with respect to requirements negotiation. However, we were not able to assess the cost of pending open issues because the projects were confined to writing an RS and did not involve its complete implementation. To measure the *Uncertainty and Equivocality Reduction*, we used the number of resolved uncertainties and ambiguities during both asynchronous discussions and negotiation meetings. To ensure the appropriate classification of issues, these measures were derived through the categorization performed by the client groups during the inspection and then checked by one of the researchers. Last, we measured the presence of *Common Ground* during the negotiation meetings by means of 1) the number of speaking turns and spoken words, 2) the number of questions, clarifications, and extra info, and 3) the number of references to posted messages written in the asynchronous discussion. Using triangulation for measuring the same construct of interest is one of the methods of mitigating threats to construct validity. Since these measures were obtained from the content analysis of the meeting transcripts, we took care of threats to construct validity by using a couple of independent coders and measuring the intercoder agreement. We could then presume that the overall threats to construct validity were minor.

External validity concerns the ability to generalize the results to industry practice. Since students played the roles of both clients and developers, they might not be representative of the population of professional stakeholders. This threat is partially mitigated by the presence of Canadian students, who were attending a specific course on GSD and were then trained on meeting protocols and negotiation techniques for requirements engineering. Some students also had previous working experience in the software industry. Furthermore, the requirements documents inspected in this study might not be a representative of industrial requirements documents in general as our

projects were Web applications. Nevertheless, the geographical distribution and cultural diversity of our groups make our study a representative of global projects with stakeholders overseas or in different time zones. One could also assume that software teams are characterized by diversity in working experience, knowledge about meeting protocols, and negotiation techniques.

5 CONCLUSIONS

Requirements engineering is essentially a collaborative endeavor that involves a large pool of stakeholders processing a significant amount of information about the problem domain and solution space. To add to the complexity of this collaborative effort, stakeholder groups handle inevitable uncertainties about the desired change in the client organization and manage different views in the problem situation. Hence, requirements engineering becomes an intensive decision-making and continuous negotiation process, significantly affected by the stakeholders' geographical distribution, which impedes colocated meetings. Although videoconferencing is regarded—thanks to its richness—as the most appropriate medium for effectively conducting distributed negotiations [14], it is associated with an additional overhead and is not very practical: Even when everything runs smoothly, it is still difficult to conduct a long-running and productive discussion during a videoconference, especially when more than a few people are involved [36]. Drawing on media-selection theories, we proposed that a combination of lean and rich media is needed to improve the effectiveness of distributed requirements negotiations. Although synchronous communication is needed for resolving ambiguities in requirements (given the presence of multiple perspectives), groups may need time to process information and sift through these issues outside the real-time meetings in a less interactive mode; thus, asynchronous communication has been suggested as a more effective medium for carefully considering issues, as well as resolving issues that involve merely gathering missing information. Our study brings supporting evidence to these research claims. We found that not only were the requirements negotiations more effective in the mixed-media communication mode, benefiting from asynchronous discussions resolving uncertainties in requirements, but also that asynchronous discussions enabled groups to anticipate the building of common ground necessary for mutual agreements in the negotiation.

These findings have practical implications with respect to the design of a computer-mediated environment for effective distributed requirements negotiations. A combination of asynchronous text-based discussions to reduce the uncertainty in requirements and synchronous videoconference meetings to converge on those requirements issues requiring equivocality reduction should form the backbone of such an environment. Moreover, the asynchronous discussions provide a structure for the discussion of requirements issues by creating a threaded discussion for each issue. During synchronous meetings, the stored threaded discussions can serve as meeting agenda and as a means for facilitating references to the threaded discussions.

Although we were not able to provide complete evidence that early asynchronous discussions can anticipate the establishment of common ground that remote stakeholders will use during videoconferencing, our findings are promising. There is a lack of research on common ground in collaborative software development, particularly in requirements engineering. With limited guidance from the software engineering literature, we believe that this study is an important starting point for the study of shared understanding and the role of computer-mediated communication in achieving it in distributed software teams. However, further studies should be conducted to obtain more confidence in the theories we put forward and the system design requirements that can be drawn from them.

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